Prova 02 - SBD

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1) a)

```
select pe.nomepes, pi.titulospil from pessoa pe, piloto pi
where pe.idpes = pi.idpes
```

b)

```
select c.idcir, c.nomecir, c.localcir from circuito c
natural join corrida c2
natural join participacao p2
where p2.voltaspar < c2.voltascor
```

c)

```
select c.nomecir, c2.datacor, c2.categoriacor from circuito c
left outer join corrida c2 on c.idcir = c2.idcir
order by c.nomecir
```

d)

```
select e.nomeequ, count(m2.idpes) as QtdeMecanicos from equipe e
natural join pessoa p
natural join mecanico m2
group by e.nomeequ
having count(m2.idpes) > 1;
```

e)

```
select p.nomepes from pessoa p
natural join piloto p2
natural join participacao p3
natural join corrida c
natural join circuito c2
where c2.nomecir = 'Interlagos'
```

2)

```
create or replace view vencedores AS
	(
		(
			select c.idcor, c.datacor, c.categoriacor, p.nomepes as nomeven,
'Piloto' as tipoven from pessoa p
			natural join piloto p2
			natural join participacao p3
			natural join corrida c
			where p3.tempopar <= all (
				select p4.tempopar from participacao p4
				natural join corrida c2
				where p3.idcor = p4.idcor
				and p3.idpes <> p4.idpes
				and p3.voltaspar=p4.voltaspar
			)

		)
	union all
		(
			select c.idcor, c.datacor, c.categoriacor, e.nomeequ as nomeven,
'Equipe' as tipoven from pessoa p
			natural join equipe e
			natural join participacao p3
			natural join corrida c
			where p3.tempopar <= all (
				select p4.tempopar from participacao p4
				natural join corrida c2
				where p3.idcor = p4.idcor
				and p3.idpes <> p4.idpes
				and p3.voltaspar=p4.voltaspar
			)
		)
	)

SELECT * FROM vencedores v ORDER BY v.tipoven, v.idcor;
```

3)

```
WITH RECURSIVE busca_mecanico_supervisor(nivel, idsup, idpes, nomeequ, nomepes,
especialidademec) AS
	(
		SELECT 0, raiz.idsup, raiz.idpes, raiz.nomeequ, raiz.nomepes,
raiz.especialidademec
		FROM (mecanico NATURAL JOIN pessoa NATURAL JOIN equipe) AS raiz

	WHERE raiz.idpes = raiz.idsup
	UNION ALL
		SELECT nivel+1, filho.idsup, filho.idpes, filho.nomeequ, filho.nomepes,
filho.especialidademec
		FROM busca_mecanico_supervisor pai, (mecanico NATURAL JOIN pessoa
NATURAL JOIN equipe) AS filho
			WHERE filho.idsup = pai.idpes
			AND filho.nomeequ=pai.nomeequ
			AND filho.idpes <> pai.idpes
		)

SELECT DISTINCT nivel, idsup, idpes, nomepes, especialidademec
FROM busca_mecanico_supervisor as bms
WHERE nomeequ = 'Toyota'
ORDER BY bms.nivel, bms.idsup, bms.idpes;
```